VISTA: Situated Visual Modality Adaptation for Spoken VLM Responses in XR

HyunA Seo
hyuna.seo@hcs.snu.ac.kr
Seoul National University
Seoul, Republic of Korea

Junhee Kim
junheek@snu.ac.kr
Seoul National University
Seoul, Republic of Korea

Seunghyeon Lee
lsh09130@gmail.com
Seoul National University
Seoul, Republic of Korea

Rajesh Balan
rajesh@smu.edu.sg
School of Information Systems
Singapore Management University
Singapore, Singapore

Youngki Lee
youngkilee@snu.ac.kr
Computer Science and Engineering
Seoul National University
Seoul, Republic of Korea

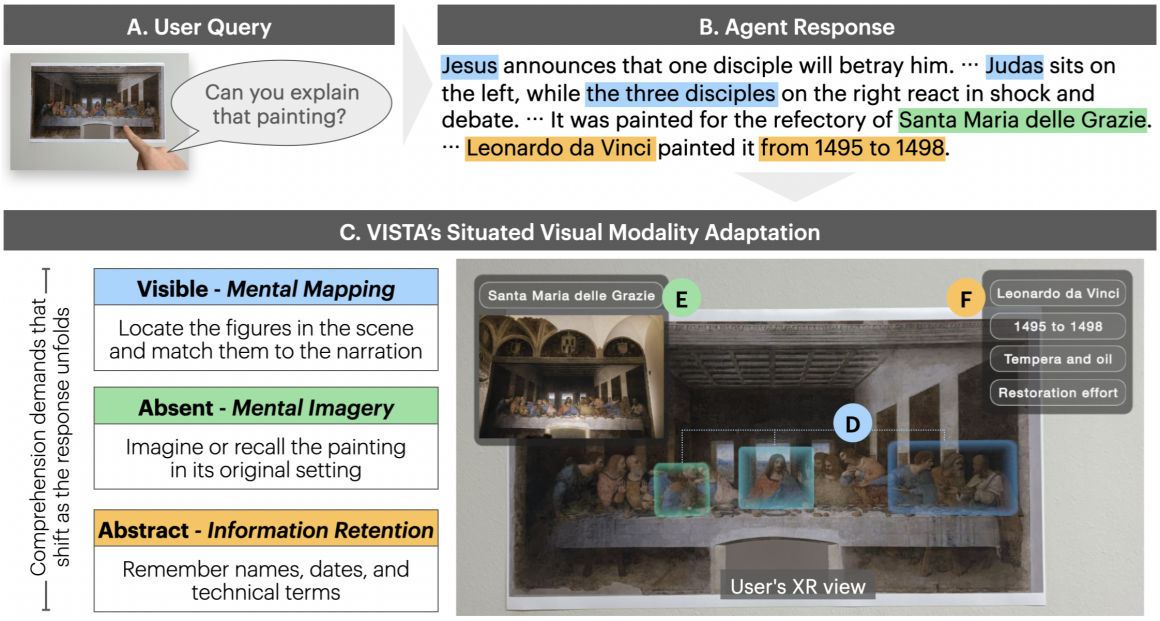


Figure 1: Example operation of VISTA. Given a user's query and egocentric view (A), the VLM generates a free-form, information-rich response (B). Situated visual modality adaptation (C) generates visual support for three comprehension demands (left) at run time. Spatial highlights (D) guide the user's gaze to each described figure, a reference image (E) shows the painting's original setting, and pinned keywords (F) keep unfamiliar names, dates, and terms in view.

ABSTRACT

We present VISTA, an XR agent system that delivers free-form, information-rich VLM responses with adaptive visual support to improve comprehension and reduce cognitive load. A response can shift among visible objects, absent content, and abstract facts, requiring users to locate, imagine, and retain information while listening. Existing XR agents, however, typically deliver every response in one fixed visual form that supports only a single demand. To address this, we present situated visual modality adaptation, an approach that (i) adapts visual support as the demand changes within a response and (ii) synchronizes each cue with the narration. The key idea is to use the response–scene relation, i.e., how the described content relates to the user's physical surroundings, to select and generate matched visual support at run time. We performed a user study and interviews with a working prototype, and participants showed improved comprehension with lower workload compared to an audio-with-text baseline.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality**; *Interactive systems and tools.*

KEYWORDS

Extended reality, XR agents, vision-language models, multimodal interaction

1 INTRODUCTION

Recent advances in VLMs are transforming how XR agents help people explore their surroundings [25, 59]. Using egocentric camera streams and additional sensors, these agents interpret the user's physical context and answer a wide range of situated questions [17, 46, 77]. For example, a shopper might ask, "What is this used for?" and a museum visitor, "Tell me about this painting." Such questions elicit detailed, free-form responses, and today's XR agents handle this interaction through a voice interface, with spoken queries and audio answers [27, 56]. Consider a visitor viewing Leonardo da Vinci's *The Last Supper* (Figure 1A). The agent would respond (Figure 1B): "Jesus announces that one disciple will betray him… It was painted for the refectory of Santa Maria delle Grazie… Leonardo da Vinci painted it from 1495 to 1498." As the narration continues, the user must map each description to the correct figure (highlighted in blue), imagine a place they cannot see (green), and retain unfamiliar details (yellow). A single information-rich response can impose all of these demands at once. Delivered as a uniform audio stream, it forces the user to resolve every demand in real time, and failing any one undermines comprehension of the rest [45, 82].

Recent work has explored augmenting verbal explanations in XR with visual cues to support comprehension. Most of this work pairs a response with a single form of visual support, serving one comprehension demand. Systems built for predefined query types (e.g., "What is this?") attach anchored labels and highlights that point users to the object in question [38, 46, 72]. Agents that answer open-ended questions also keep one form for every response, whether a text transcript [27, 39, 56], an object-anchored panel [23], or embodied gestures [8, 51]. A few systems adapt their cues to the type of information, showing a highlight for a location, an arrow for a motion, and a gesture for a spatial relation [51, 73, 86]. Even with this adaptation, the support stays within a single demand, typically helping users locate and follow visible targets (Figure 2A). In contrast, free-form, information-rich responses are improvisational. Preparing visual support in advance is difficult, as the content of a response emerges only after generation [18, 21, 50] and the type of support it needs changes as the narration progresses (Figure 2B).

We present Vista, an XR agent system that delivers free-form, information-rich VLM responses as audio narration paired with adaptive visual support to improve comprehension while reducing cognitive load. It employs situated visual modality adaptation, an approach that matches visual support to the comprehension demands of each part of a response. Understanding such a response involves three demands: (i) mental mapping to locate each described target in the physical scene [37, 75], (ii) mental imagery to imagine or recall described content absent from the user's view [11, 62], and (iii) information retention to hold abstract details in memory as the narration moves on [45, 57]. Vista generates the matched cues at run time and presents each one just in time with the narration, aiming to improve comprehension while reducing cognitive load.

We distinguish three response–scene relations by how the described content relates to the user's current physical environment: *Visible*, *Absent*, and *Abstract*. Returning to the Figure 1 example, our approach generates visual support on the fly and overlays it onto the user's view (C). For Visible content, spatial highlights (D) direct the user's attention to each described figure, removing the burden of searching the scene. For Absent content, a reference image (E) externalizes the original setting the user would otherwise have to imagine. For Abstract content, pinned keywords (F) hold unfamiliar names, dates, and terms in view, helping the user retain and revisit them. Vista segments the response by these relations and provides an overview UI, so the user can grasp the scope of the response and pace its delivery.

We demonstrate Vista across five everyday scenarios: art explanation, presentation Q&A, machine setup, product comparison, and matcha brewing. Videos of all five scenarios are available at https://vista-xr.github.io/. We evaluated each stage of the automated pipeline on 30 egocentric image–query pairs spanning six categories. We also conducted a user study with 12 users in the art explanation and beverage crafting scenarios to verify Vista 's effectiveness compared to the audio-with-text baseline. Participants showed improved comprehension in both objective (20.3%) and self-reported measures (29.9%), and significantly lower perceived workload (22.3%) under Vista's integrated delivery. Our contributions are as follows:

- We introduce situated visual modality adaptation, an approach that transforms free-form agent responses into adaptive visual support at run time based on how each response segment relates to the user's physical environment.
- We design and implement Vista, an XR agent system that automatically segments free-form responses, generates matched visual support, and synchronizes each cue with the audio narration at run time.
- We demonstrate through a within-subjects user study ($N = 12$) that Vista improves comprehension and reduces perceived workload compared with a standard audio-with-text baseline.

2 RELATED WORK

2.1 Visual Support for XR Agent Responses

Recent XR agents use VLMs to answer a wide range of questions about the user's surroundings [33, 77, 85], and researchers have proposed visual cues that help users understand responses delivered through audio narration. Most of this work serves a single comprehension demand (Figure 2A). Systems built for predefined query types (e.g., "What is this?" or "Where did I leave my car keys?") support them with concise visual cues, such as anchored labels and highlights that point to the queried object [29, 38, 40, 46, 72]. Agents that answer open-ended questions instead use uniform visual support independent of the response content, whether a text transcript [27, 39, 56, 77], simplified text [84], an object-anchored panel [23], or embodied expressions [8]. Fewer systems adapt the visual support to the information at hand, building on a task or a set of objects known in advance [47, 73]. Guided Reality selects among highlights, arrows, and hand motions to guide each step of a machine task [86], and AgentHands generates deictic, affective (e.g., a thumbs-up), and iconic (e.g., numbers) hand gestures for pre-registered objects to support understanding [51].

In contrast, a single free-form, information-rich response can direct users to visible objects, describe absent places or states, and mention names or values worth retaining, so the dominant demand itself shifts as the narration progresses (Figure 2B). The responses

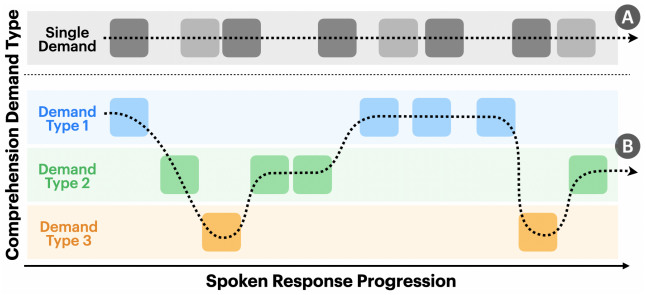


Figure 2: Prior work assumes a single comprehension demand per response (A). In a free-form, information-rich response, the dominant demand changes as the narration progresses (B), so no single form of visual support serves the whole response.

we target are improvisational. What a response will contain becomes known only after it is generated, so the visual modality must be selected across demands at run time. Our approach adapts visual support to how each described content relates to the user's physical context, addressing each demand as it arises.

2.2 In-Situ Visual Augmentation

XR has long overlaid virtual content onto physical surroundings to convey situated information [10, 28]. Prior systems have used this capability in two broad ways: (i) directing attention to relevant information already present and (ii) externalizing information that is not currently visible. The first is useful when physical surroundings are cluttered or unfamiliar: the attention funnel draws a tunnel of virtual frames from the user's view to a target [6]; highlights and bounding boxes outline or tint relevant objects and regions [36, 68, 74]; and arrows indicate locations, directions, or movements [14, 71]. These cues have supported physical tasks: classic systems overlaid assembly instructions directly on parts [76], ProcessAR authors in-situ 2D/3D instructions from expert demonstrations [16], and surgical telementoring anchors annotations in the operating field [24].

The second externalizes what the user cannot currently see, including absent objects, unfamiliar places, and future or target states. Such information has been shown as 2D reference images, 3D models, or 360° panoramas. Earlier systems prepared this content for specific sites or scenarios, from prerecorded panoramas [15] to manually authored historical reconstructions [58, 61]. Recent systems retrieve or generate it on demand: RealityTalk [48] augments live presentations with visuals triggered by the presenter's speech, RealityChat [35] retrieves reference images and maps during live conversations, and Thing2Reality [34] turns 2D content into 3D objects that participants can manipulate in XR meetings. XR thus offers a wide spectrum of visual support, from a simple box drawn over an object to a panorama that transforms the surrounding view. Making a free-form agent response easy to follow requires placing the right cue in the right place at the right moment, based on how each piece of information relates to the user's physical environment.

2.3 Audio-Visual Coordination

Humans process visual and auditory information through separate channels, so pairing visuals with audio narration can effectively deliver XR agent responses [26, 53, 60, 69]. However, simply attaching visual cues to speech without precise temporal alignment can split attention and add cognitive load.

Two factors make coordination especially critical in XR. First, the narrow field of view (FoV) limits how much content can be displayed simultaneously, so users must focus their attention deliberately [49, 84]. Second, when audio and visuals arrive out of sync, users must hold one in working memory while waiting for the other, increasing time pressure and consuming cognitive resources [2, 13, 41, 63]. AgentHands [51] shows that precisely timing gestures to speech improves agent communication, but coordinating diverse visual modalities within a single response remains underexplored.

These principles were established with authored instructional media, where the content, its visuals, and their timing are fixed in advance [54]. An XR agent's response, in contrast, is generated only at run time, and its described targets are scattered across a scene that differs with every question. Whether and how the principles carry over to such delivery remains an open question. VISTA operationalizes them at run time, aligning each visual cue with the corresponding moment in the narration so that users immediately see what is being discussed, grounding verbal descriptions in their view.

3 VISTA DESIGN

We aim to automatically transform the free-form, information-rich responses of XR agents into audio-visual presentations, with two design goals: (i) improving comprehension and (ii) reducing cognitive load. Visual support could be adapted based on response-level properties such as length, structure, or topic. These properties characterize a response as a whole and do not indicate how comprehension demands change from one part to the next. Within a single response, however, the focus continually shifts among described targets, including entities, states, and facts. Each part imposes a comprehension demand based on how its content relates to the user's physical surroundings. We capture this as the **response–scene relation** and use it as an intermediate representation to transform each part of a free-form response into appropriate visual support at run time. We distinguish three relations by the demand each imposes (Section 3.1) and design visual modalities that satisfy each relation's requirement (Section 3.2).

3.1 Response–Scene Relations

We assign a response–scene relation to each response part using two distinctions: (i) whether the described content appears in the user's current view and (ii) whether it has a useful visual counterpart. Content in view is *Visible*. Content out of view is *Absent* when it has a visual counterpart and *Abstract* when it has none. We next unpack the comprehension demand imposed by each response–scene relation and derive the corresponding requirement for visual support.

Visible. When the described target is in the user's view, the primary demand is **mental mapping**. As the agent describes a tool or region in front of them, the user must search the scene to locate

that target and connect it to the narration [37, 75]. This search costs more time and attention in cluttered or unfamiliar environments [3, 74, 83], and repeated searches across targets cause the user to fall behind the narration. To ease this burden, visual support should guide the user's gaze to each target as it is mentioned.

Absent. When the described content is not present in the user's current physical environment, the primary demand is **mental imagery** [11, 62]. For example, the response could describe a tool stored in another room, the fully grown state of a visible plant, or the nighttime appearance of a place seen during the day. No search of the current scene can reveal such content, so the user must imagine or recall it while following the narration [87, 88]. Building this image occupies visuospatial working memory [70], so visual support should externalize the missing entity or state in a form that matches the description.

Abstract. Some information has no useful visual counterpart but remains **difficult to retain**. Unfamiliar terms, precise values, and proper names easily vanish when delivered through speech alone [45, 57]. Displaying the full response as text would keep these details accessible, but it also adds reading burden in XR [67, 81]. Visual support should instead keep only essential details in view as brief keywords that users can revisit throughout the narration.

3.2 Visual Modality Mapping

We realize each relation's requirement through visual modalities, concrete forms of visual support that reduce the cognitive load each demand imposes. For each response–scene relation, we first derive a base modality that serves its primary demand, and then introduce extended modalities for needs the base alone does not cover.

Base modalities. Figure 3 illustrates the mapping from each response–scene relation to its base modality.

- **Visible: Spatial Highlight.** We render a spatial highlight as a semi-transparent box anchored to each visible target, guiding the user's gaze without obscuring the underlying physical object (Figure 3A, B). When the response refers to multiple visible targets, each highlight receives a distinct color so the user can distinguish the targets as the narration moves among them.
- **Absent: Reference Image.** We externalize the absent content with a reference image. When image search can provide a suitable reference, we retrieve one, such as the view from the building's rooftop (Figure 3D). When the described state is specific to an object in the user's scene, a retrieved image would depict a different instance, so we generate an image that renders the state on the user's object, such as the finished texture of the matcha in their own bowl (Figure 3C).
- **Abstract: Keyword.** We extract retention-critical details, such as exact terms, dates, values, and proper nouns, and pin them in the user's view as concise keywords (Figure 3E, F). The keywords remain visible throughout the narration, allowing the user to revisit them after they are mentioned.

Extended modalities. Base modalities address the primary demand of each response–scene relation. Some content also describes a motion to follow, a place to experience, a form to inspect, or details to retain. We fulfill these additional needs with extended modalities that augment the base. Figure 4 shows the extended modalities for visible (top) and absent (bottom) content.

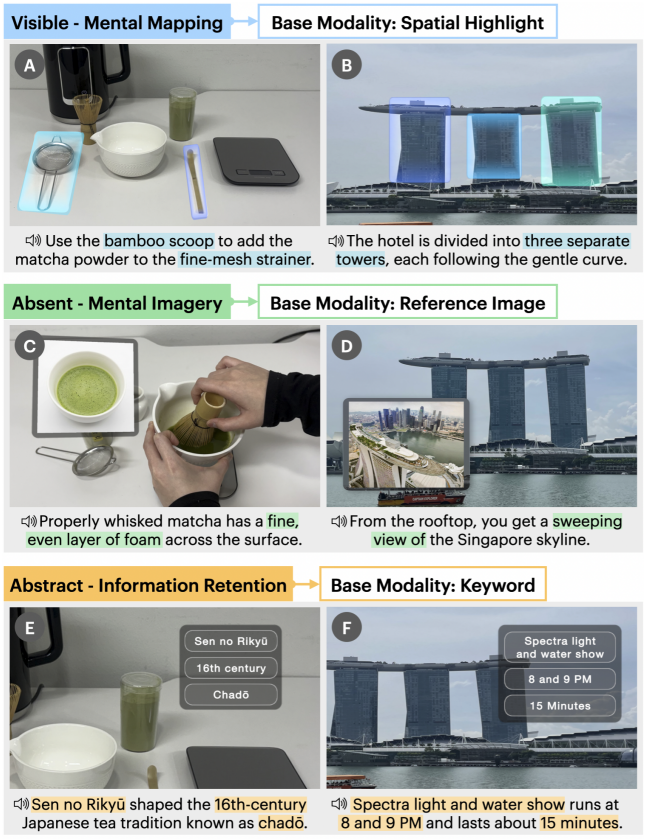


Figure 3: Examples of base modalities for visible, absent, and abstract content in matcha preparation (left) and sightseeing (right). Narration shown below each panel.

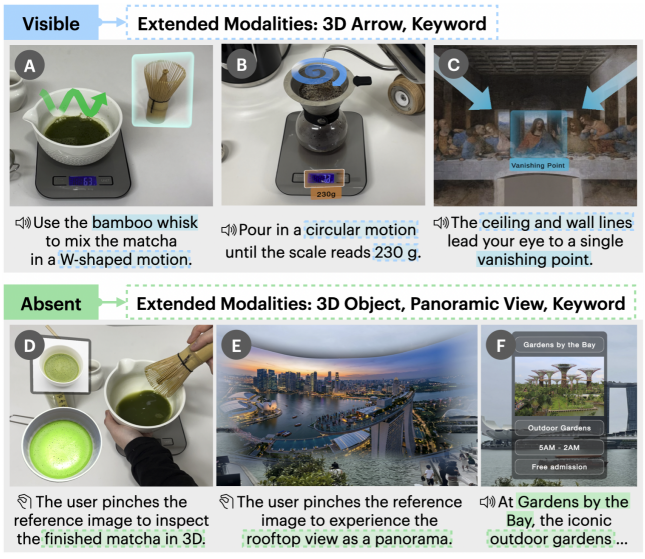


Figure 4: Extended modalities for Visible content (A–C) and Absent content (D–F).

- **3D Arrow for Directionality.** A spatial highlight marks where a target is but cannot express how it should move [65], leaving the user to infer the trajectory from speech alone. When the response describes a motion or a direction, such as the W-shaped whisking of the matcha (Figure 4A), a concentric spiral pour (B), or perspective lines converging toward a painting's vanishing point (C), a 3D arrow traces it in the scene.
- **3D Object for Interactive Inspection.** A reference image externalizes absent content but shows it from a fixed viewpoint and scale [43]. When closer inspection from different viewpoints or scales is useful, we provide an interactive 3D object. Keeping the object visible throughout the narration could occlude the scene, so it remains hidden until the user chooses to open it. For example, pinching the reference image of the finished matcha in Figure 3C opens the 3D object shown in Figure 4D. The user can then rotate, resize, and position it anywhere in the scene to inspect the foam and texture more closely.
- **Panoramic View for Spatial Context.** A flat reference image shows what a place looks like but cannot convey its surrounding spatial context. We generate a panoramic view of the described environment and render it around the user, providing an immersive experience of the place (Figure 4E) [20, 52]. The panorama remains hidden until the user opens it by pinching the reference image (e.g., Figure 3D).
- **Keyword for Retention.** Visible and Absent content can also impose retention demands. A weight reading mentioned alongside a visible scale (Figure 4B) or the name and visiting details of a place shown in a reference image (F) can be difficult to retain as the narration continues. We supplement the base modality with keywords whenever such details appear.

4 VISTA WALKTHROUGH

This walkthrough shows how VISTA realizes situated visual modality adaptation. A user wearing an XR headset stands before Leonardo da Vinci's The Last Supper and asks, "Can you tell me about this?" The system answers in two run-time phases. Phase 1 converts the user's situated query into a navigable response overview (Figure 5). Phase 2 prepares and delivers the synchronized audio-visual presentation for each segment the user chooses to explore (Figure 6).

Phase 1: From Query to Overview. Once the system receives the egocentric image stream and speech query (1), the Response Generator 2 produces a detailed response grounded in the user's view. In parallel, Ack Generator 3 immediately streams a short spoken acknowledgment ("Sure, let me look into it.") so the interaction remains conversational while the main pipeline runs. The Response Segmenter 4 then decomposes the response into semantically coherent segments, assigns each a keyword and a response–scene-relation label, and generates an overview containing the segment count, the main target, the keywords, and a one-sentence gist. For instance, the *Artist and Location* segment is labeled *Absent* because it describes the painting's original setting, which is not present in the user's physical space. Then, the Object Detector 5 localizes the painting for overview placement, while Text-to-Speech 6 synthesizes the gist audio. Once both outputs are ready, the Sync Manager 7 anchors the overview UI. It plays the spoken gist, allowing the user to inspect the response structure before selecting a segment to explore in detail.

Phase 2: Per-Segment Delivery. As soon as segmentation finishes, all segments enter the Modality Assigner 8 for background preparation. Here we trace Segment 0, which is labeled *Absent*. Following the mapping in Section 3.2, the Modality Assigner selects *reference image* as the base modality. It also adds *Panoramic View* as an extended modality for deeper spatial context and extracts three retention-critical *keywords*. It then rewrites the segment for natural spoken delivery.

Visual and audio assets are generated in parallel. The Visual Modality Generator 9 routes each specification to the corresponding handler. For example, it fetches the refectory photograph, generates an immersive panoramic view, and extracts keywords. Meanwhile, the Audio Modality Generator 10 inserts a timing mark at each point in the narration where a visual cue should be rendered in the XR view. It also adds an audio pause mark (e.g., 750 ms) to give the user time to process the visual information. Text-to-Speech then synthesizes the narration and returns the timestamp.

When the user selects the segment from the overview UI, the Sync Manager 11 begins playback. The three keywords and the reference image first appear in a semi-transparent standby state. As the narration progresses, each keyword activates at its first mention. When the narration reaches 'Santa,' the refectory image appears, and the narration briefly pauses. At the end of playback,

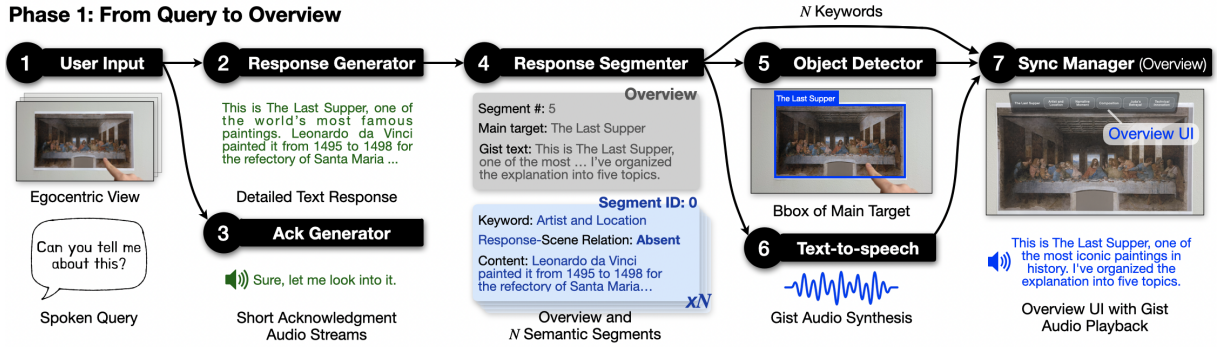


Figure 5: Phase 1 of the VISTA walkthrough. The pipeline turns a user query into a segmented response and presents a spatially anchored overview with gist audio.

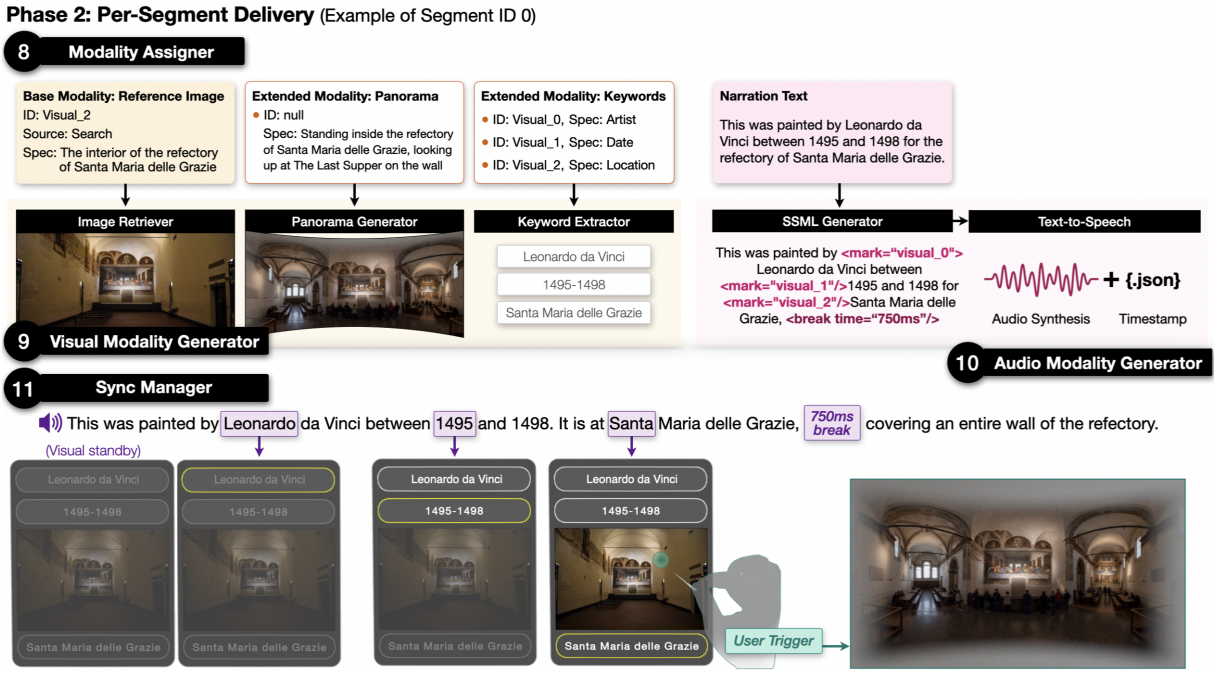


Figure 6: Phase 2 of the walkthrough. Using Segment 0 as an example, the pipeline assigns visual modalities, generates visual and audio assets in parallel, and synchronizes their delivery.

the user can pinch the reference image to open the panoramic view. This interaction pattern enables Vista to support immediate comprehension during the narrated explanation while reserving more visually disruptive expansions for deliberate user control.

5 IMPLEMENTATION

5.1 System Overview

We implement Vista as a run-time response-delivery system for XR agents following the architecture shown in Figure 7. The XR client runs in Unity 6000.1.17f1 on an Android-based Samsung Galaxy XR headset.[1] It samples egocentric camera frames at 1 fps and sends them with the user's speech query to the server over WebSocket. The server executes the pipeline described in Section 4, including response generation, segmentation, modality assignment, and asset generation. Playback remains on the client, where the Sync Manager activates each visual cue when the narration reaches the corresponding content.

Two scheduling choices reduce idle waiting. (i) The client receives the overview and its spoken gist as soon as segmentation is complete. (ii) The user can inspect the response structure through the overview UI while the server prepares the segment assets. The server processes segments in parallel and generates their visual and audio assets concurrently. Each segment becomes playable when its own assets are ready.

To realize this delivery, we divide each response into segments that carry a single response–scene relation (Section 5.1.1), synchronize each visual cue with the moment its content is narrated

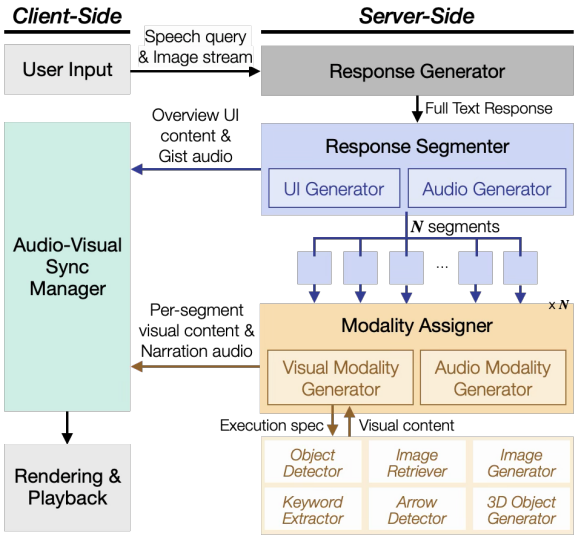


Figure 7: System overview of Vista.

(Section 5.1.2), and provide an overview through which the user explores the response at their own pace (Section 5.1.3). Appendices A and B detail the APIs, prompts, and output formats.

5.1.1 Response Segmentation. A free-form response must be segmented so that visual support can change with its response–scene relation. The structure a VLM already generates does not provide reliable boundaries. A single recipe step, for example, can describe cutting the visible vegetables, the target cut shape, and the exact amounts, spanning all three relations. Assigning one modality to

[1] https://www.samsung.com/us/xr/galaxy-xr/

such content leaves some demands unsupported, and showing several modalities together makes the user alternate among mental processes, adding switch costs [31, 80] and visual clutter. Segmenting every sentence avoids the mixing but fragments closely related information.

We set boundaries using transitions of the response–scene relation and breaks in semantic coherence. The segmenter processes the response one sentence at a time, identifying each sentence's main target and assigning its response–scene relation by the two distinctions in Section 3.1. It places a boundary whenever the relation changes or the semantic connection to the preceding sentence breaks. A step may thus be split when it mixes relations, and consecutive sentences that share a relation and remain coherent stay together. Each segment then maps to one base modality.

5.1.2 Audio-Visual Synchronization. The user must connect each visual cue to the corresponding part of the narration. Without temporal alignment, this matching becomes a demand in itself. We coordinate the two channels in time, exploiting the human capacity to process auditory and visual information in parallel [53, 55]. Two principles guide this coordination: (1) every visual cue is triggered at the first mention of its content, and (2) the narration's tempo adapts to the response–scene relation, since modalities differ in the time they take to process.

The Audio Modality Generator inserts a timing mark immediately before the first mention of each visual element. It also adds pauses based on the visual modality. For Visible content, spatial highlights and arrow lead-ins receive a 300 ms pause. This interval accommodates gaze shifts, as a saccade with fixation takes roughly 250 ms [64]. For Absent content, reference images receive a 750 ms pause for initial inspection [32, 64]. Keywords add no pause as they repeat the spoken content.

All cues first appear in a semi-transparent standby state. Each cue turns fully opaque as the narration reaches it, and returns to a de-emphasized state once the narration moves past. This incremental reveal limits visual overload and maintains temporal alignment between the two channels. Figure 8 illustrates this delivery for a Visible segment and Figure 6 ⓫ shows an Absent case.

5.1.3 Overview-Based Navigation. Continuous playback does not reveal the response structure or let the user jump to a specific segment. We present an overview with selectable segments so users can inspect a response's scope and choose where to begin. The layout stays consistent across responses of different lengths, topics, and structures.

When segmentation completes, the system presents the overview with a one-sentence spoken gist. It anchors the overview near the main target while preserving its visibility [78]. Each segment appears as a button labeled with a keyword. Selecting a button starts the segment's synchronized audio-visual delivery, and previously viewed segments remain available for replay.

5.2 Pipeline Evaluation

We conducted a stage-level evaluation to examine how the pipeline performs across diverse situations and queries. Our goal was to characterize the automated response-delivery pipeline across a broader range of situations than the live user study could cover.

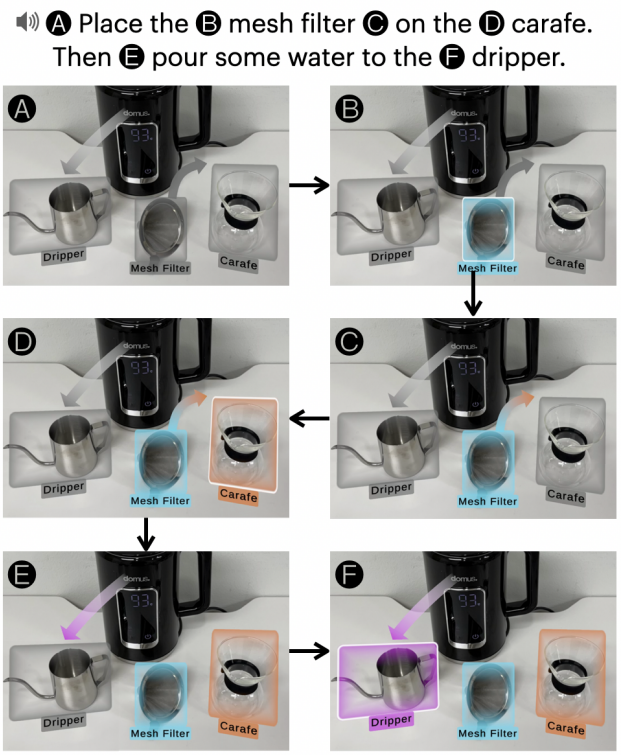


Figure 8: Audio-visual synchronization in a Visible segment. Timing marks (A–F) inserted in the narration activate each cue at the first mention of its content: cues start in a semi-transparent standby state and turn fully opaque when mentioned.

We therefore evaluated each stage using fixed single-frame inputs, each consisting of one egocentric image and one query. Fixed inputs make the analysis reproducible and isolate errors in response segmentation, modality assignment, and visual and audio generation from errors introduced by head motion, depth sensing, or rendering. We collected 30 egocentric image–query pairs spanning six everyday categories, including cooking, art, shopping, and home repair. Two authors independently judged each output as correct or incorrect against a shared set of criteria, and another author adjudicated the items on which they disagreed. As each downstream stage depends on the correctness of the preceding one, we report conditional stage accuracy: each stage was evaluated only on segments that passed all earlier stages. Visual modalities are selectively invoked based on response content and modality requirements, making a single end-to-end accuracy difficult to interpret across different execution paths. Conditional evaluation instead characterizes the reliability of each stage when it is invoked. Appendix C provides the full scenario set and judgment criteria.

Results. The 30 scenarios produced 161 segments (M=5.4, SD=1.2, range 4–8); Table 1 reports the accuracy of each stage. Segmentation failed when a single sentence mixed visible and absent content, such as one describing the jacket on the rack while also referring to how it drapes when worn.[2]

[2]All six segments that failed here were labeled absent. Following the conditional protocol, we exclude them from the Modality Assigner counts.

Table 1: System accuracy evaluated on 161 segments from 30 scenarios. Each stage is evaluated only on segments that passed all preceding stages. Rows shaded in light blue report the accuracy of the Modality Assigner ❽; rows shaded in light gray report the accuracy of the Visual Modality Generator ❾.

Stage	Correct / Total	Acc.
Response Segmenter ❹	155 / 161	96.3%
Modality Assigner ❽		
Visible	110 / 113	97.3%
Spatial highlight	94 / 110	85.5%
3D arrow	27 / 32	84.4%
Keywords	92 / 95	96.8%
Absent	30 / 30	100%
Reference Image	25 / 31	80.6%
⌞ Search	8 / 10	80.0%
⌞ Generated	17 / 21	81.0%
Panoramic view	8 / 10	80.0%
3D object	14 / 19	73.7%
Keywords	23 / 24	95.8%
Abstract	12 / 12	100%
Audio Modality Generator ❿	134 / 152	88.2%

The Modality Assigner was almost always correct; accuracy dropped on the generation side. Failures for visible content were ones of coverage: a highlight marked only some of the objects a segment mentioned (e.g., four of six eggs), or an arrow's endpoint was detected in the wrong place. For absent content, the visual showed the right object or place but differed in detail: retrieved images placed the target in the wrong context (e.g., anemone species instead of how an anemone feeds), and generated images and 3D models missed the target state (e.g., a pouch sewn from a fabric other than the user's).

Audio generation for synchronization failed when the SSML markup ordered the cues incorrectly, omitted the pause before a reference image, or carried some timing marks that did not match the number of visual elements they were meant to trigger.

6 USER EVALUATION

We conducted a within-subjects study to evaluate whether Vista improves comprehension and reduces perceived workload compared with an audio-with-text baseline. The study was approved by the university's Institutional Review Board.

6.1 Study Design

Baseline Design. We implemented an audio-with-text interface modeled after the built-in Gemini assistant of the Samsung Galaxy XR headset [27]. The agent narrates the full response, and a persistent panel provides playback controls (i.e., pause, resume, and replay) and an expandable text transcript. Both conditions share the same query input channels, response generation model, prompt, and TTS engine. They differed only in response delivery. The baseline delivered the response as one continuous narration. Vista divided the generated response into segments by response–scene relation



Figure 9: Response delivery timeline, averaged over the 24 user-study queries per condition. The x-axis is time after the end of the user's query.

and synchronized visual cues with the narration. This comparison isolates the combined effect of Vista's response-delivery design.

Condition Comparability. Figure 9 shows the delivery timeline measured over the 24 study queries per condition. Acknowledgment onset (Baseline: 4.97 vs. Vista: 4.67 s) and main audio onset (16.90 vs. 16.64 s) did not differ between conditions (both $p > .14$). The main audio was also comparable in total duration (97.3 vs. 98.7 s) and word count (227 vs. 211 words; both $p > .10$). In Vista, segment preparation overlapped with the acknowledgment, gist, and overview exploration: on average, the first segment became playable at 30.7 s, before participants' first selection at 38.0 s. This rules out response latency and the amount of narrated content as explanations for the differences we report below.

Procedure and Tasks. After a brief tutorial on the headset's hand tracking and voice interaction, participants completed one declarative task and one procedural task in each condition, for four tasks in total (Figure 10). Both task types elicited long, information-dense responses grounded in the participant's physical surroundings. In the declarative task, participants stood before a printed reproduction of Leonardo da Vinci's The Last Supper or the Korean royal screen painting Irworobongdo and asked the agent to explain it. In the procedural task, they followed the agent's instructions to prepare pour-over coffee or matcha tea. We organized the tasks into two matched sets, and each condition used a different set. We counterbalanced condition order and set to reduce content-specific learning effects. Participants completed a comprehension quiz and questionnaires after each task. After all four tasks, we conducted a 15-minute semi-structured interview. Each session lasted approximately 90 minutes.

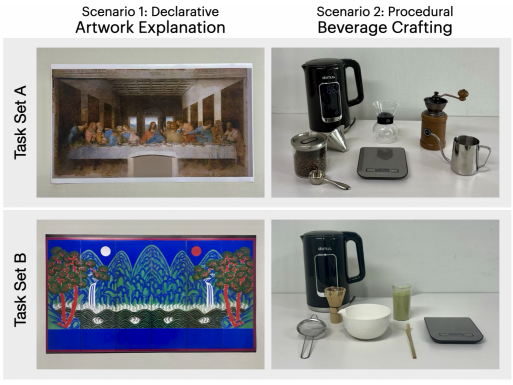


Figure 10: User evaluation tasks. Each participant completed all four tasks. Task sets A and B contain matched content, and each condition used a different set.

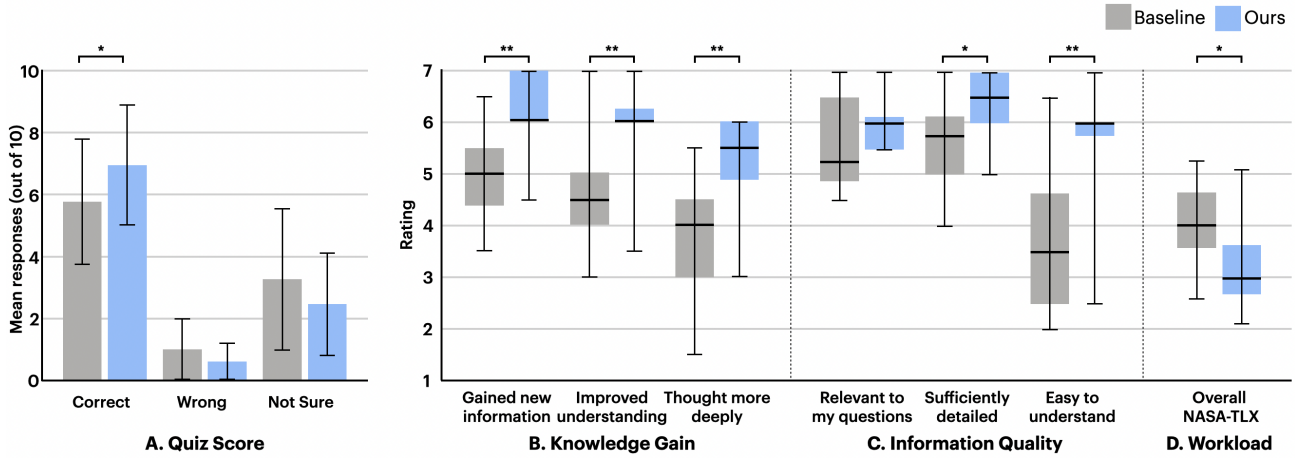


Figure 11: Quantitative results. (A) Comprehension quiz responses. (B) Perceived knowledge gain. (C) Perceived information quality. (D) Overall NASA-TLX workload. Bars in A show means with standard deviations. Boxes in B, C, and D show medians and interquartile ranges. Higher values indicate better in A, B, and C. Lower values indicate lower workload in D. Brackets mark significant differences between conditions based on Holm-adjusted p-values (* $p < .05$, ** $p < .01$, *** $p < .001$).

Participants. We recruited 12 participants (4 female, 8 male; mean age = 24.58, SD = 3.00) from our institution via word of mouth. All participants had prior XR experience. Each participant was compensated US$20 for their time.

Measurements. We measured objective comprehension with a 10-item, four-choice *comprehension quiz* administered after each task [42]. Responses were generated at run time, so the system generated each quiz from the response received by that participant. All quizzes used the same prompt, item count, and format (see Appendix D). We instructed participants to leave items blank when unsure. To assess subjective information acquisition, we measured *perceived knowledge gain* with three cognitive learning outcome items [66] and *perceived information quality* through ratings of relevance, detail, and clarity [1, 22].

We measured perceived workload with NASA-TLX [30] as an indicator of cognitive load. We also assessed overall usability with SUS [4, 12]. The semi-structured interviews examined per-task impressions and cross-condition comparisons.

Data Collection and Analysis. After each task, participants completed a post-task questionnaire. Unless otherwise noted, questionnaire items used 7-point Likert scales. We compared conditions using Wilcoxon signed-rank tests and reported Holm-adjusted p-values. Quiz accuracy and overall NASA-TLX workload were the primary outcomes, representing the study's comprehension and cognitive-load effects. All remaining measures were exploratory. We applied Holm–Bonferroni correction across the two primary outcomes and within each exploratory scale family. Full test statistics and matched-pairs rank-biserial correlations (r) are provided in Appendix E. We analyzed the interview data using reflexive thematic analysis [9]. One author coded the transcripts inductively, and the authors refined the themes through discussion.

6.2 Results

Participants achieved higher comprehension scores and reported lower perceived workload with Vista than with the baseline. We first report results from the comprehension quiz and questionnaires, followed by interview findings that describe how participants experienced the two response-delivery conditions.

6.2.1 Quantitative Results. Figure 11 shows that participants using Vista achieved higher comprehension scores and reported greater knowledge gain with lower perceived workload than in the baseline condition.

Response Comprehension. Participants answered significantly more quiz questions correctly with Vista ($M = 6.92$, $SD = 1.93$) than with the baseline ($M = 5.75$, $SD = 2.01$; $p = .048$, $r = .67$). They also rated all three perceived knowledge gain items higher with Vista (all $p \leq .008$). Detail ($p = .020$) and clarity ratings ($p = .009$) were higher with Vista. As expected, relevance ratings did not differ between conditions ($p = .125$), since both conditions used matched tasks and the same response-generation setup.

Perceived Workload. Participants reported lower overall NASA-TLX workload with Vista ($M = 3.14$, $SD = 0.83$) than with the baseline ($M = 4.04$, $SD = 0.79$; $p = .032$, $r = -.77$). Mental demand ($p = .006$), temporal demand ($p = .021$), and frustration ($p = .020$) were lower with Vista. Participants also rated their task performance higher ($p = .006$).

Usability. The mean SUS scores were 74.9 ($SD = 14.7$) for Vista and 69.7 ($SD = 16.9$) for the baseline, and both exceeded the acceptability threshold of 68 [5].

6.2.2 Qualitative Findings. Semi-structured interviews revealed how different aspects of Vista shaped participants' experiences of receiving XR agent responses.

Information in each response–scene relation was easier to follow. In the baseline, participants handled mental mapping, imagery, and retention on their own while the narration continued. Vista resolved the demand each response–scene relation imposed and improved comprehension. We describe what participants experienced in each relation below.

- **Visible.** Participants located visible targets without searching the scene on their own (10/12). In the baseline, participants searched

while the narration continued and sometimes missed what followed (3/12). Since Vista spatially highlighted each target as it was mentioned, P7 could put that effort into understanding the response. P4 described this experience as *"joint attention with the agent"* and compared it to *"following a kind guide."* Similarly, in the baseline, participants inferred the intended motion and the direction to look from the narration alone. In our system, 3D arrows showed both directly and helped them interpret the narration more accurately (5/12).

- **Absent.** Our system externalized absent content so participants could see and understand what the narration described (8/12). In the baseline, participants looked around to check whether the described content was in the scene when the response shifted to absent content (5/12). They sometimes lost track of the narration or returned to the transcript (3/12). With Vista, participants could inspect absent content in reference images while the narration continued. P2 could linger over information that *"would have slipped past as audio."* All participants also explored the panoramas and 3D objects. P8 said that manipulating 3D objects helped them understand unfamiliar objects and tasks more intuitively, and P3 described the panoramas as immersive and said they made the explanation more memorable.
- **Abstract.** Participants retained unfamiliar terms, proper nouns, and quantities throughout the response while their attention stayed on the physical scene (10/12). In the baseline, participants often missed these details in the audio, and their attention repeatedly shifted from the physical scene to the transcript (2/12). With ours, anchored keywords made complex names and numbers easier for P2 to follow and helped P5 recall the content at a glance after the narration ended.

Taken together, Vista addressed the comprehension demand associated with each response–scene relation and adapted its support as the response moved from one relation to another. In the baseline, participants had to determine how to follow the response as its demands changed, which distracted them from the narration (5/12). Vista made this judgment for them by identifying each segment's dominant demand and adapting the presentation accordingly. This reduced the effort of shifting among mental mapping, imagery, and retention (6/12), a process that prior work has shown to impose measurable cognitive costs [31, 80]. Participants could focus on the narration and develop a deeper understanding of the agent's response. They described the experience as offloading the effort of acquiring information onto the agent (P7), work they had to *"figure out myself"* in the baseline (P4).

Temporal synchronization reduced attentional overhead and increased trust. Participants explicitly appreciated that each visual cue appeared when the corresponding content was narrated (10/12). Some said this increased trust in the agent (4/12), while others said it freed attention for deeper understanding (3/12). P1 described it as *"like an expert standing right beside me, pointing to each thing at exactly the right moment."* P6 added that synchronized cues reduced the need to manually connect spoken information to the physical world and left them *"more confident answering the quiz."*

The overview UI gave users more control over pacing. Participants valued the overview UI as they could (i) scan the response topics before playback (9/12) and (ii) choose a starting point when listening again (7/12). Specifically, participants used the overview as a progress tracker in procedural tasks (*"I can see the five steps. Just do these and I'm done!"*, P10). This suggests that the overview helped users traverse the explanation more actively rather than receiving it as a single forced stream.

7 DISCUSSION

Vista improved comprehension and reduced workload as an integrated system. Our findings suggest that an agent answering open-ended questions must coordinate its delivery at run time by dividing each response, choosing a visual form for each part, and timing each cue as the narration unfolds. The following subsections discuss future directions raised by participant feedback and limitations of our approach.

7.1 Toward User-Aware Response Delivery

Participants differed in where comprehension broke down. P6 retained numerical details well, so the keywords added little for them. Spatial highlights helped them locate referents among unfamiliar tools. P1 struggled to picture spaces from speech, and panoramas clarified the scale and layout of those spaces. They found 3D objects less valuable. These patterns are consistent with prior findings that visual search, working-memory capacity, and mental imagery ability differ across individuals [7, 19, 44, 79].

Our approach selects support from the response and its relation to the physical scene. This decision rule provided a useful starting point across the individual bottlenecks observed in our study without requiring a user model. The interviews also revealed needs that response content alone could not indicate. P1, who was unfamiliar with art terminology, found the overview keywords too detailed to choose a starting segment. Future systems could refine the initial selection using user-side signals such as prior knowledge, interaction history, and the desired amount of explanation. These signals could adjust keyword detail, access to extended modalities, and the overview structure.

7.2 Balancing Automation and User Control

Vista automated the selection of visual support and its synchronization with the narration, reducing the user's comprehension burden. Participants controlled pacing, order, and level of detail through the overview and optional visuals (i.e., panoramic views and 3D objects). This division suggests a practical starting point for automation in XR agents. Since Vista cannot know how quickly a user comprehends an explanation or how much detail they need, users should remain able to adjust these aspects.

Future work could lower the cost of exercising this control. Adjusting the delivery currently requires manual actions, such as opening the overview or pinching a reference image. Participants who rarely used XR (P2, P8) or who felt more comfortable speaking found these manipulations effortful (P5). Lighter mechanisms, such as short voice commands for tempo (e.g., "next" to play the following segment) and depth (e.g., "tell me more about that"), could make these adjustments more immediate and intuitive. Complementary signals from XR sensing offer another strategy. Gaze dwelling on a highlight, repeated fixations on a keyword, or a shift away from the target region could suggest that a segment needs more time or

less detail. The system could act on these signals provisionally, and the user would retain the final decision.

7.3 Calibrating Trust in Visual Support

Participants used spatial and temporal alignment to judge the agent's reliability. P1 reported that highlights landing on the correct object confirmed that their understanding matched the agent's intent, and four participants associated synchronized cues with greater trust. P3 also imagined using reference images to verify ingredient identity in future cooking scenarios. This benefit, however, carries a risk. In our pipeline evaluation, reference image generation reached 80.6% accuracy and 3D arrows 84.4%. Though these accuracies reflect the current state of the underlying generative models, an inaccurate cue could reinforce a wrong belief if users read visual support as implicit confirmation. P7 raised this concern, noting that misplaced trust during furniture assembly could lead to irreversible mistakes. Future systems should help users calibrate trust in visual support through confidence indicators, source attribution, or controls to correct or dismiss inaccurate cues.

7.4 Granularity of Response–Scene Relations

Our three response–scene-relation types provide an initial representation of how response content relates to the physical scene. This granularity offered a useful basis for selecting support in our study. Future systems could model these relations more precisely. One direction is to decompose the Absent type. Absent content may describe an entity located elsewhere, a past or future state of a visible object, or another instance of a familiar category. These cases differ in their spatial, temporal, and instance-level relations to the scene. Future systems could model each dimension separately. Along the temporal dimension, a future state of the user's bowl could be overlaid directly on the bowl, an object could be animated to show how its state changes over time, and a slider could let users inspect its prior states. These representations could connect absent states more closely to the physical objects they describe.

8 CONCLUSION

We present Vista, an XR agent system that delivers free-form, information-rich VLM responses as synchronized audio-visual presentations. Our method segments each response based on the response–scene relation of its content and adaptively generates matched visual support at runtime. We also synchronize each visual cue with the moment its content is narrated and provide an overview that lets users navigate the response at their own pace. We verify Vista 's effectiveness by evaluating a fully working prototype in a user study, showing that it provides higher comprehension and lower perceived workload than an audio-with-text baseline.

REFERENCES

[1] Vassilios P. Aggelidis and Prodromos D. Chatzoglou. 2012. Hospital information systems: Measuring end user computing satisfaction (EUCS). *J. Biomed. Inform.* 45, 3 (2012), 566–579.

[2] Faisal M Alessa, Mohammed H Alhaag, Ibrahim M Al-Harkan, Mohamed Z Ramadan, and Fahad M Alqahtani. 2023. A neurophysiological evaluation of cognitive load during augmented reality interactions in various industrial maintenance and assembly tasks. *Sensors* 23, 18 (2023), 7698.

[3] Huidong Bai, Prasanth Sasikumar, Jing Yang, and Mark Billinghurst. 2020. A user study on mixed reality remote collaboration with eye gaze and hand gesture sharing. In *Proc. CHI*. 1–13.

[4] Aaron Bangor, Philip T Kortum, and James T Miller. 2008. An empirical evaluation of the system usability scale. *Int. J. Hum.-Comput. Interact.* 24, 6 (2008), 574–594.

[5] Aaron Bangor, Philip T. Kortum, and James T. Miller. 2009. Determining What Individual SUS Scores Mean: Adding an Adjective Rating Scale. *J. Usability Stud.* 4, 3 (2009), 114–123.

[6] Frank Biocca, Arthur Tang, Charles Owen, and Fan Xiao. 2006. Attention funnel: omnidirectional 3D cursor for mobile augmented reality platforms. In *Proc. CHI*. 1115–1122.

[7] Olessia Blajenkova, Maria Kozhevnikov, and Michael A. Motes. 2006. Object-spatial imagery: A new self-report imagery questionnaire. *Appl. Cogn. Psychol.* 20, 2 (2006), 239–263.

[8] Riccardo Bovo, Steven Abreu, Karan Ahuja, Eric J Gonzalez, Li-Te Cheng, and Mar Gonzalez-Franco. 2025. Embardiment: an embodied ai agent for productivity in xr. In *Proc. IEEE VR*. IEEE, 708–717.

[9] Virginia Braun and Victoria Clarke. 2019. Reflecting on reflexive thematic analysis. *Qual. Res. Sport Exerc. Health* 11, 4 (2019), 589–597.

[10] Nathalie Bressa, Henrik Korsgaard, Aurélien Tabard, Steven Houben, and Jo Vermeulen. 2021. What's the situation with situated visualization? A survey and perspectives on situatedness. *IEEE TVCG* 28, 1 (2021), 107–117.

[11] Robert Eamon Briscoe. 2018. Superimposed Mental Imagery: On the Uses of Make-Perceive. In *Perceptual Imagination and Perceptual Memory*, Fiona Macpherson and Fabian Dorsch (Eds.). Oxford University Press, 161–185.

[12] J Brooke. 1996. SUS: A quick and dirty usability scale. *Usability Evaluation in Industry* (1996).

[13] Josef Buchner, Katja Buntins, and Michael Kerres. 2022. The impact of augmented reality on cognitive load and performance: A systematic review. *J. Comput. Assist. Learn.* 38, 1 (2022), 285–303.

[14] Yuanzhi Cao, Anna Fuste, and Valentin Heun. 2022. Mobiletutar: A lightweight augmented reality tutorial system using spatially situated human segmentation videos. In *Proc. CHI EA*. 1–8.

[15] Shenchang Eric Chen. 1995. QuickTime VR: An Image-Based Approach to Virtual Environment Navigation. In *Proc. SIGGRAPH*. 29–38.

[16] Subramanian Chidambaram, Hank Huang, Fengming He, Xun Qian, Ana M Villanueva, Thomas S Redick, Wolfgang Stuerzlinger, and Karthik Ramani. 2021. Processar: An augmented reality-based tool to create in-situ procedural 2d/3d ar instructions. In *Proc. DIS*. 234–249.

[17] Hyunsung Cho, Jacqui Fashimpaur, Naveen Sendhilnathan, Jonathan Browder, David Lindlbauer, Tanya R Jonker, and Kashyap Todi. 2025. Persistent assistant: Seamless everyday AI interactions via intent grounding and multimodal feedback. In *Proc. CHI*. 1–19.

[18] Christopher Chou, Lisa Dunlap, Koki Mashita, Krishna Mandal, Trevor Darrell, Ion Stoica, Joseph E. Gonzalez, and Wei-Lin Chiang. 2025. VisionArena: 230K Real World User-VLM Conversations with Preference Labels. In *Proc. CVPR*. 3877–3887.

[19] Alasdair D. F. Clarke, Jessica L. Irons, Warren James, Andrew B. Leber, and Amelia R. Hunt. 2022. Stable individual differences in strategies within, but not between, visual search tasks. *Q. J. Exp. Psychol.* 75, 2 (2022), 289–296.

[20] James J. Cummings and Jeremy N. Bailenson. 2016. How Immersive Is Enough? A Meta-Analysis of the Effect of Immersive Technology on User Presence. *Media Psychol.* 19, 2 (2016), 272–309.

[21] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven C. H. Hoi. 2023. InstructBLIP: Towards General-Purpose Vision-Language Models with Instruction Tuning. In *NeurIPS*, Vol. 36. 49250–49267.

[22] William H. DeLone and Ephraim R. McLean. 2003. The DeLone and McLean model of information systems success: A ten-year update. *J. Manag. Inf. Syst.* 19, 4 (2003), 9–30.

[23] Mustafa Doga Dogan, Eric J Gonzalez, Karan Ahuja, Ruofei Du, Andrea Colaço, Johnny Lee, Mar Gonzalez-Franco, and David Kim. 2024. Augmented object intelligence with xr-objects. In *Proc. UIST*. 1–15.

[24] Danilo Gasques, Janet G Johnson, Tommy Sharkey, Yuanyuan Feng, Ru Wang, Zhuoqun Robin Xu, Enrique Zavala, Yifei Zhang, Wanze Xie, Xinming Zhang, et al. 2021. Artemis: A collaborative mixed-reality system for immersive surgical telementoring. In *Proc. CHI*. 1–14.

[25] Gemini Team, Google. 2026. Introducing Gemini 3 Flash: Benchmarks, Global Availability. https://blog.google/products/gemini/gemini-3-flash/.

[26] Paul Ginns. 2006. Integrating information: A meta-analysis of the spatial contiguity and temporal contiguity effects. *Learn. Instr.* 16, 6 (2006), 511–525.

[27] Google. 2024. Gemini Live for Galaxy XR. https://blog.google/products/gemini/. Accessed: 2026.

[28] Jens Grubert, Tobias Langlotz, Stefanie Zollmann, and Holger Regenbrecht. 2016. Towards pervasive augmented reality: Context-awareness in augmented reality. *IEEE TVCG* 23, 6 (2016), 1706–1724.

[29] Aditya Gunturu, Shivesh Singh Jadon, Nandi Zhang, Morteza Faraji, Jarin Thundathil, Wesley Willett, and Ryo Suzuki. 2025. RealitySummary: Exploring On-Demand Mixed Reality Text Summarization and Question Answering using Large Language Models. In *Proc. SUI*. Association for Computing Machinery, 21:1–21:13.

[30] Sandra G Hart. 2006. NASA-task load index (NASA-TLX); 20 years later. In *Proc. HFES*, Vol. 50. Sage publications Sage CA: Los Angeles, CA, 904–908.

[31] Caro Hautekiet, Sam Verschooren, Naomi Langerock, and Evie Vergauwe. 2023. Attentional Switching Between Perception and Memory: Examining Asymmetrical Switch Costs. *Atten. Percept. Psychophys.* 85, 5 (2023), 1471–1489.

[32] John M. Henderson. 2003. Human Gaze Control During Real-World Scene Perception. *Trends Cogn. Sci.* 7, 11 (2003), 498–504.

[33] Teresa Hirzle, Florian Müller, Fiona Draxler, Martin Schmitz, Pascal Knierim, and Kasper Hornbæk. 2023. When XR and AI Meet: A Scoping Review on Extended Reality and Artificial Intelligence. In *Proc. CHI*. 1–22.

[34] Erzhen Hu, Alex Olwal, Mingyi Li, David Kim, Jungtaek Hong, Seongkook Heo, Xun Qian, and Ruofei Du. 2025. Thing2Reality: Enabling Spontaneous Creation of 3D Objects from 2D Content using Generative AI in XR Meetings. In *Proc. UIST*.

[35] Shivesh Jadon, Mehrad Faridan, Edward Mah, Rajan Vaish, Wesley Willett, and Ryo Suzuki. 2024. Augmented conversation with embedded speech-driven on-the-fly referencing in AR. *arXiv preprint arXiv:2405.18537* (2024).

[36] Denis Kalkofen, Erick Mendez, and Dieter Schmalstieg. 2007. Interactive Focus and Context Visualization for Augmented Reality. In *Proc. ISMAR*.

[37] Yuki Kamide, Gerry T. M. Altmann, and Sarah L. Haywood. 2003. The Time-Course of Prediction in Incremental Sentence Processing: Evidence from Anticipatory Eye Movements. *J. Mem. Lang.* 49, 1 (2003), 133–156.

[38] Yoonsang Kim, Devshree Jadeja, Divyansh Pradhan, Yalong Yang, and Arie E. Kaufman. 2026. SpeechLess: Micro-utterance with Personalized Spatial Memory-aware Assistant in Everyday Augmented Reality. In *Proc. IEEE VR*. IEEE, 217–227.

[39] Yoonsang Kim, Divyansh Pradhan, Devshree Jadeja, and Arie E. Kaufman. 2026. From Speech-to-Spatial: Grounding Utterances on A Live Shared View with Augmented Reality. In *Proc. IEEE VR*. IEEE, 228–238.

[40] Yoonsang Kim, Yalong Yang, and Arie E. Kaufman. 2026. Memento: Towards Proactive Visualization of Everyday Memories with Personal Wearable AR Assistant. In *Proc. IEEE VRW*. IEEE, 993–1000.

[41] You-Jin Kim, Radha Kumaran, Jingjing Luo, Tom Bullock, Barry Giesbrecht, and Tobias Höllerer. 2025. On the Go with AR: Attention to Virtual and Physical Targets while Varying Augmentation Density. In *Proc. CHI*. 1–16.

[42] Walter Kintsch. 1994. Text comprehension, memory, and learning. *Am. Psychol.* 49, 4 (1994), 294–303.

[43] Uri Korisky and Liad Mudrik. 2021. Dimensions of Perception: 3D Real-Life Objects Are More Readily Detected Than Their 2D Images. *Psychol. Sci.* 32, 10 (2021), 1636–1648.

[44] Maria Kozhevnikov, Stephen Kosslyn, and Jennifer Shephard. 2005. Spatial versus object visualizers: A new characterization of visual cognitive style. *Mem. Cognit.* 33, 4 (2005), 710–726.

[45] Wayne Leahy and John Sweller. 2011. Cognitive load theory, modality of presentation and the transient information effect. *Appl. Cogn. Psychol.* 25, 6 (2011), 943–951.

[46] Jaewook Lee, Jun Wang, Elizabeth Brown, Liam Chu, Sebastian S Rodriguez, and Jon E Froehlich. 2024. GazePointAR: A context-aware multimodal voice assistant for pronoun disambiguation in wearable augmented reality. In *Proc. CHI*. 1–20.

[47] Chenyi Li, Guande Wu, Gromit Yeuk-Yin Chan, Dishita Gdi Turakhia, Sonia Castelo Quispe, Dong Li, Leslie Welch, Claudio Silva, and Jing Qian. 2025. Satori 悟り: Towards Proactive AR Assistant with Belief-Desire-Intention User Modeling. In *Proc. CHI*. 1–24.

[48] Jian Liao, Adnan Karim, Shivesh Singh Jadon, Rubaiat Habib Kazi, and Ryo Suzuki. 2022. RealityTalk: Real-time speech-driven augmented presentation for AR live storytelling. In *Proc. UIST*. 1–12.

[49] David Lindlbauer, Anna Maria Feit, and Otmar Hilliges. 2019. Context-aware online adaptation of mixed reality interfaces. In *Proc. UIST*. 147–160.

[50] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023. Visual Instruction Tuning. In *NeurIPS*, Vol. 36. 34892–34916.

[51] Ziyi Liu, David Li, Zhongyi Zhou, David Kim, Ruofei Du, and Xun Qian. 2026. AgentHands: Generating Interactive Hands Gestures for Spatially Grounded Agent Conversations in XR. In *Proc. CHI (CHI)*. ACM.

[52] Guido Makransky and Gustav B. Petersen. 2021. The Cognitive Affective Model of Immersive Learning (CAMIL): A Theoretical Research-Based Model of Learning in Immersive Virtual Reality. *Educ. Psychol. Rev.* 33, 3 (2021), 937–958.

[53] Richard E Mayer. 2002. Multimedia learning. In *Psychol. Learn. Motiv.* Vol. 41. Elsevier, 85–139.

[54] Richard E. Mayer. 2005. *The Cambridge Handbook of Multimedia Learning*. Cambridge University Press.

[55] Richard E. Mayer. 2014. Temporal Contiguity Principle. In *The Cambridge Handbook of Multimedia Learning* (2nd ed.). Cambridge University Press, 153–167.

[56] Meta. 2023. Ray-Ban Meta Smart Glasses. https://www.meta.com/smart-glasses/. Accessed: 2026.

[57] George A. Miller. 1956. The Magical Number Seven, Plus or Minus Two: Some Limits on Our Capacity for Processing Information. *Psychol. Rev.* 63, 2 (1956), 81–97.

[58] Eslam Nofal, Ahmed M. Elhanafi, Hendrik Hameeuw, and Andrew Vande Moere. 2018. Architectural Contextualization of Heritage Museum Artifacts Using Augmented Reality. *Stud. Digit. Herit.* 2, 1 (2018), 42–67.

[59] OpenAI. 2026. GPT-5.6: Frontier Intelligence that Scales with Your Ambition. https://openai.com/index/gpt-5-6/. Accessed: 2026.

[60] Allan Paivio, James M Clark, et al. 2006. Dual coding theory and education. *Pathways to literacy achievement for high poverty children* 1 (2006), 149–210.

[61] George Papagiannakis, Sebastien Schertenleib, Brian O'Kennedy, Marlene Arevalo-Poizat, Nadia Magnenat-Thalmann, Andrew Stoddart, and Daniel Thalmann. 2005. Mixing virtual and real scenes in the site of ancient Pompeii. *Comput. Animat. Virtual Worlds* 16, 1 (2005), 11–24.

[62] Joel Pearson. 2019. The Human Imagination: The Cognitive Neuroscience of Visual Mental Imagery. *Nat. Rev. Neurosci.* 20, 10 (2019), 624–634.

[63] Shwetha Rajaram, Hemant Bhaskar Surale, Codie McConkey, Carine Rognon, Hrim Mehta, Michael Glueck, and Christopher Collins. 2025. Gesture and audio-haptic guidance techniques to direct conversations with intelligent voice interfaces. In *Proc. CHI*. 1–20.

[64] Keith Rayner. 1998. Eye Movements in Reading and Information Processing: 20 Years of Research. *Psychol. Bull.* 124, 3 (1998), 372–422.

[65] Patrick Renner and Thies Pfeiffer. 2017. Evaluation of Attention Guiding Techniques for Augmented Reality-based Assistance in Picking and Assembly Tasks. In *Proc. IUI Companion*. ACM, 89–92.

[66] Alfred P. Rovai, Mervyn J. Wighting, Jason D. Baker, and Linda D. Grooms. 2009. Development of an instrument to measure perceived cognitive, affective, and psychomotor learning in traditional and virtual classroom higher education settings. *Internet High. Educ.* 12, 1 (2009), 7–13.

[67] Rufat Rzayev, Paweł W. Woźniak, Tilman Dingler, and Niels Henze. 2018. Reading on Smart Glasses: The Effect of Text Position, Presentation Type and Walking. In *Proc. CHI*. Association for Computing Machinery, 1–9.

[68] Björn Schwerdtfeger and Gudrun Klinker. 2008. Supporting Order Picking with Augmented Reality. In *Proc. ISMAR*.

[69] Roland Sigrist, Georg Rauter, Robert Riener, and Peter Wolf. 2013. Augmented visual, auditory, haptic, and multimodal feedback in motor learning: a review. *Psychon. Bull. Rev.* 20, 1 (2013), 21–53.

[70] Valerie K. Sims and Mary Hegarty. 1997. Mental Animation in the Visuospatial Sketchpad: Evidence from Dual-Task Studies. *Mem. Cognit.* 25, 3 (1997), 321–332.

[71] Marco Speicher, Christoph Rosenberg, Donald Degraen, Florian Daiber, and Antonio Krúger. 2019. Exploring visual guidance in 360-degree videos. In *Proc. ACM TVX*. 1–12.

[72] Sruti Srinidhi, Edward Lu, and Anthony Rowe. 2024. Xair: An xr platform that integrates large language models with the physical world. In *Proc. ISMAR*. IEEE, 759–767.

[73] Daniel Stover and Doug Bowman. 2024. TAGGAR: General-Purpose Task Guidance from Natural Language in Augmented Reality using Vision-Language Models. In *Proc. SUI*. 1–12.

[74] Jonathan Sutton, Tobias Langlotz, Alexander Plopski, Stefanie Zollmann, Yuta Itoh, and Holger Regenbrecht. 2022. Look over there! investigating saliency modulation for visual guidance with augmented reality glasses. In *Proc. UIST*. 1–15.

[75] Michael K. Tanenhaus, Michael J. Spivey-Knowlton, Kathleen M. Eberhard, and Julie C. Sedivy. 1995. Integration of Visual and Linguistic Information in Spoken Language Comprehension. *Science* 268, 5217 (1995), 1632–1634.

[76] Arthur Tang, Charles Owen, Frank Biocca, and Weimin Mou. 2003. Comparative effectiveness of augmented reality in object assembly. In *Proc. CHI*. 73–80.

[77] Yiliu Tang, Jason Situ, Andrea Yaoyun Cui, Mengke Wu, and Yun Huang. 2025. LLM Integration in Extended Reality: A Comprehensive Review of Current Trends, Challenges, and Future Perspectives. In *Proc. CHI*. 1–24.

[78] Hsin-Ruey Tsai, Shih-Kang Chiu, and Bryan Wang. 2025. GazeNoter: Co-Piloted AR Note-Taking via Gaze Selection of LLM Suggestions to Match Users' Intentions. In *Proc. CHI*. 1–18.

[79] Nash Unsworth and Randall W. Engle. 2007. The nature of individual differences in working memory capacity: Active maintenance in primary memory and controlled search from secondary memory. *Psychol. Rev.* 114, 1 (2007), 104–132.

[80] Sam Verschooren, Sebastian Schindler, Rudi De Raedt, and Gilles Pourtois. 2019. Switching Attention from Internal to External Information Processing: A Review of the Literature and Empirical Support of the Resource Sharing Account. *Psychon. Bull. Rev.* 26, 2 (2019), 468–490.

[81] Matthew D. Ward and William S. Helton. 2022. Dual-Task Interference while Receiving Information on a Head Mounted Display and Manual Tracking with and without Auditory Warnings. *Appl. Ergon.* 101 (2022), 103713.

[82] Christopher D. Wickens. 2002. Multiple resources and performance prediction. *Theor. Issues Ergon. Sci.* 3, 2 (2002), 159–177.

[83] Jeremy M. Wolfe and Todd S. Horowitz. 2017. Five Factors that Guide Attention in Visual Search. *Nat. Hum. Behav.* 1, 3 (2017), 0058.

[84] Guande Wu, Jing Qian, Sonia Castelo Quispe, Shaoyu Chen, João Rulff, and Claudio Silva. 2024. Artist: Automated text simplification for task guidance in augmented reality. In *Proc. CHI*. 1–24.

[85] Yufeng Wu, Raj Mahmud, Abdullah Bin Sawad, Elise van den Hoven, and Baki Kocaballi. 2026. Conversational Agents in Augmented Reality: A Systematic Review. *ACM TiiS* (2026). Just Accepted.
[86] Ada Yi Zhao, Aditya Gunturu, Ellen Yi-Luen Do, and Ryo Suzuki. 2025. Guided reality: Generating visually-enriched ar task guidance with llms and vision models. In *Proc. UIST*. 1–15.
[87] Rolf A. Zwaan. 2016. Situation Models, Mental Simulations, and Abstract Concepts in Discourse Comprehension. *Psychon. Bull. Rev.* 23, 4 (2016), 1028–1034.
[88] Rolf A. Zwaan and Gabriel A. Radvansky. 1998. Situation Models in Language Comprehension and Memory. *Psychol. Bull.* 123, 2 (1998), 162–185.

A ADDITIONAL IMPLEMENTATION DETAILS

Table 2 summarizes the run-time models, services, and client components that complement the pipeline description in the main paper. The prompt instructions and output schemas for segmentation, modality assignment, and cue generation are provided in Section B.1.

Table 2: Stage-level implementation details corresponding to the system walkthrough in the main paper.

Stage	Implementation
❶ Input	Samsung Galaxy XR; Camera2 egocentric capture at 1 fps; OpenXR hand input; server-side voice activity detection.
❷ Response Generator	Gemini 3 Flash conditioned on the speech query and latest egocentric frame.
❸ Ack Generator	Gemini 2.5 Flash Native Audio for immediate spoken acknowledgment.
❹ Response Segmenter	Prompted VLM with Pydantic-validated structured output for segments, keywords, response–scene-relation labels, and overview metadata.
❺ Object Detector	VLM-based target detection returning normalized bounding boxes.
❻ Gist TTS	Google Cloud TTS for overview narration.
❼ Overview Delivery	Unity `Physics.Raycast` for world-space anchoring and a client-side Sync Manager for coordinated UI and gist playback.
❽ Modality Assigner	Prompted VLM with Pydantic-validated cue assignments, execution specifications, and spoken narration.
❾ Visual Modality Generator	VLM-based localization and arrow planning; SerpApi and Gemini 3.1 Flash for reference imagery; Meshy6 and GLTFast for 3D assets.
❿ Audio Modality Generator	SSML timing-mark generation and Google Cloud TTS with millisecond-level mark timestamps.
⓫ Segment Playback	Unity-based Sync Manager that activates each visual element using its corresponding TTS timestamp.

B SYSTEM PROMPTS AND SCHEMAS

All stages use Pydantic[3]-validated JSON schemas. Below, we show the core instructions for each stage. Prompts and schemas are reproduced verbatim from the implementation; the schema field `referent_availability` and the prompt term "referent state" correspond to the response–scene relation label used in the paper.

B.1 Response Segmenter 4

You are given: (1) the user's query, (2) the full response text, and (3) the user's egocentric images. Decompose the response into N semantic segments. Place a boundary when the dominant referent changes or when it shifts between something the user can see and something they cannot. Do not mirror the response's own section structure; one original section may need splitting if it mixes visible and non-visible referents.

Per segment, output:

- keyword: Extract a concise 1–3 word topic label that summarizes this segment.
- content: Copy the verbatim text span from the original response.
- availability_reasoning: Think step-by-step:
  (1) Identify the main target of this segment. What entity, region, or concept does it primarily talk about?
  (2) Is the main target conceptual, numerical, or abstract — something with no physical form that could be pointed at? If yes → abstract. Stop here.
  (3) Look at the user's egocentric images. Can you see the main target? If it is not visible → absent. Stop here.
  (4) The target is visible. Would highlighting it help the user locate or distinguish it in the scene? If yes → visible.
- referent_availability: The label you arrived at above.

Global output:

- segment_count: How many segments you produced.
- main_target: The single most prominent object or region that the entire response is about. This will be used to anchor the overview UI in the user's view.
- gist_text: Write a one-sentence spoken summary that naturally includes the segment count (e.g., "This is The Last Supper, and I can explain it through 5 main topics.").
- core_keyword: Extract a 2–3 word label for the entire response.

B.2 Modality Assigner 8

You are given: (1) the segment content, (2) its referent_availability label from Response Segmenter, and (3) the user's egocentric images.

Read the segment and list every element that deserves its own visual cue. All elements in a segment share the same referent state. For each element, reason through the following:

Per-element reasoning:

(1) Modality. The referent state determines the base modality: visible → spatial_highlight (the user can see it; highlight where it is), absent → reference_image (the user cannot see it; show what it looks like), abstract → keyword (there is nothing to point at; pin the term on screen). For present and absent segments, also identify terms, values, or proper nouns that the listener would struggle to retain from audio alone and assign them as additional keyword elements.
(2) Extended modality. Does this element describe a motion or directional relationship between two targets? → add 3D_arrow. Does it reference a place the user cannot physically visit but would benefit from experiencing spatially? → add panoramic_view. Does the user need to compare the referent's physical scale or shape against a real counterpart? → add 3D_object.
(3) Execution spec. Describe the target in enough detail for a downstream handler to act on. Be specific: "the fourth person from the left at the table" for a spatial highlight, "the refectory of Santa Maria delle Grazie with The Last Supper visible on the far wall" for a reference image, "date" or "artist name" for a keyword.

Per-segment output:

- base_modality: spatial_highlight | reference_image | keyword.
- elements: Collect all elements into a flat, ordered list with sequential IDs: visual_0, visual_1, visual_2, ... Keywords are listed as their own elements alongside spatial highlights or reference images, each with its own ID. For each element, include: its type (spatial_highlight, reference_image, or keyword), its execution spec, and any extended modality flags. For Abstract segments, all elements are keywords.
- narration: Rewrite the segment text for spoken delivery. Replace redundant object names with deictic references that pair with the visual cues (e.g., "this figure right here," "as you can see").

B.3 Visual Modality Generator 9

The Visual Modality Generator invokes the following handlers to generate the visual cues selected by the Modality Assigner.

B.3.1 Spatial Highlight Handler.

You are given: (1) one or more object descriptions from the execution spec and (2) the user's egocentric image. Detect the specified objects in the image and return a bounding box for each. If multiple object names are provided (comma-separated), detect each one independently.

[3] https://docs.pydantic.dev/

Rules:

(1) Return exactly one bounding box per requested object. If multiple instances of the same object are visible, select the clearest one.
(2) Only detect objects that exactly match the requested names. Do not hallucinate or detect similar but unrequested objects.
(3) If a requested object is not visible in the image, omit it from the output.

Per-object output:

- box_2d: [ymin, xmin, ymax, xmax] normalized to $[0, 1000]$.
- label: the exact object name from the request.
- visibility_quality: a float in $[0, 1]$ rating how well the object is visible.
  - 1.0 = fully visible, clear, well-lit, unoccluded.
  - 0.7–0.9 = mostly visible with minor occlusion or lighting issues.
  - 0.4–0.6 = partially occluded or unclear.
  - 0.0–0.3 = heavily occluded, cut off at edges, or barely visible.
- quality_reasoning: brief justification for the quality score (e.g., "fully visible," "hand blocking left side," "cut off at top edge").

B.3.2 3D Arrow Handler.

You are given: (1) the execution spec from Modality Assigner and (2) the user's egocentric image.
The client has a pre-defined library of arrow primitives. Your job is to select the right combination so the arrow conveys the described motion or direction.

Output the following for each arrow:

(1) topology: Where does the arrow start and end? target-to-target (both endpoints are objects in the image), target-to-midair (starts at an object, ends in empty space), or midair-to-target (starts in empty space, ends at an object).
(2) path_style: What shape should the path take? Choose from straight, arc, loop, spiral, zigzag.
(3) flow_speed: How fast should the flow animation move? slow, normal, or fast.
(4) start_anchor, end_anchor: Locate the start and end points in the image as normalized $[x, y]$ coordinates.
(5) midair_params (only if one endpoint is mid-air): Specify the direction the arrow extends into (e.g., down, upright, forward) and the length in meters (e.g., 0.1 m).

B.4 Audio Modality Generator ⑩

You are given: (1) the narration text and (2) the ordered list of visual elements with their IDs from Modality Assigner.
Insert timing marks into the narration so the client knows exactly when to trigger each visual cue during audio playback.

Rules:

(1) Find where each element is first mentioned in the narration. Insert <mark name="<id>"/> (e.g., visual_0, visual_1) immediately before that mention.
(2) For 3D arrows, insert <mark name="arrow_start"/> before the action clause where the arrow animation should begin.
(3) Insert <break> after each mark, scaled to the element's type: 300 ms for spatial highlights (the user's gaze needs to saccade to the new target), 750 ms for reference images (the user encounters unfamiliar content for the first time), 300 ms for arrow lead-ins (the arrow appears just before the clause), and no pause for keywords (they echo what is spoken).
(4) The number of marks must exactly match the number of elements. Use sequential IDs and do not skip any.

Example (Present, three spatial highlights):

```
<speak><mark name="visual_0"/>Jesus forms a
stable triangular shape,<break time="300ms"/>
while the apostles are arranged in
<mark name="visual_1"/>the first,
<break time="300ms"/>
and <mark name="visual_2"/>third groups.</speak>
```

Table 3: The 30 scenarios used for the pipeline evaluation, grouped by category. Each row shows the scene description followed by the user query.

#	Scene / Query
Art & Culture	
1	Single Western painting on the wall. → Tell me about this painting.
2	East Asian calligraphy or ink painting. → Can you interpret this for me?
3	Sculpture or installation art. → What is this?
4	Historic building exterior. → What's the history behind this building?
Food & Cooking	
5	Pour-over coffee setup on counter. → How do I brew pour-over with this?
6	Vegetables and seasonings around cutting board. → What can I make with these?
7	Side dishes on the table. → Tell me about these side dishes.
8	Baking ingredients on table. → I want to make macarons. How do I start?
9	Spice jars and bottles on shelf. → What are these spices, and how should I use them?
10	Fridge with food. → What can I make for dinner with what's in here?
Shopping	
11	Convenience store beverage shelf. → Can you recommend a zero-calorie drink?
12	Ramyeon display. → Which ones aren't spicy?
13	Wine shelf. → Which one goes well with steak?
14	Street with restaurant signs. → Any good restaurants around here?
Home Repair	
15	Shelf parts and hardware on floor. → How do I put this shelf together?
16	Candle jar with candle care tools. → How do I use these candle tools?
17	Sewing machine with fabric. → How do I make a pouch with this?
18	Bicycle flipped upside down. → How do I change this tire?
19	Printer showing error on display. → How do I fix this?
Home & Personal	
20	Balcony with multiple potted plants. → How do I take care of these plants?
21	Multiple medications on table. → How should I take these?
22	Photo of items to pack. → I have too much stuff. What should I leave out?
23	Home gym equipment in living room. → Can you put together a workout routine with these?
24	Car dashboard with warning lights. → What do these warning lights mean?
25	Living room with mixed objects. → Is there anything dangerous here for a toddler?
Learning & Interpretation	
26	Tide pool close-up. → What are these creatures?
27	Textbook diagram (circuit or cell). → Can you explain this diagram?
28	Map of hiking trail. → Which trail is best for beginners?
29	Board game set up mid-play. → How do you play this?
30	Presentation slide on screen. → Can you explain this slide in more detail?

C PIPELINE EVALUATION SCENARIOS AND JUDGMENT CRITERIA

To identify everyday scenarios in which users would want rich, multimodal responses from an XR agent, we conducted a formative workshop ($N = 6$). XR glasses are not yet widely adopted, so we started from participants' existing VLM usage to identify when people request information-dense responses about their real-world surroundings. We built a local web viewer that runs on each participant's own device to keep their data private (Figure 12a–d). Before attending, participants exported their conversation histories from ChatGPT, Gemini, or both. They loaded these into the web viewer, browsed their past queries, and selected cases where they had asked about something physically in front of them and received a detailed, multi-part response. From these cases, six recurring categories emerged: Art & Culture, Food & Cooking, Shopping, Home Repair, Home & Personal, and Learning & Interpretation. For each category, we authored four to six scenarios grounded in specific real-world settings, yielding 30 evaluation scenarios in total.

C.1 Judgment Criteria

Two authors independently judged each pipeline output as correct or incorrect against the criteria below, and a third author adjudicated the items on which they disagreed. The criteria instantiate the response–scene-relation definitions in Section 3 of the main paper.

- **Response Segmenter.** A segment is correct when its boundaries follow response–scene-relation transitions or breaks in semantic coherence, and its label matches the actual relation between the segment's main target and the egocentric scene. A conditional or future state that cannot be seen in the current view is Absent, not Visible, even when it results from acting on a visible object.
- **Modality Assigner.** The assigned base and extended visual modalities must match the mapping in Section 3.2 of the main

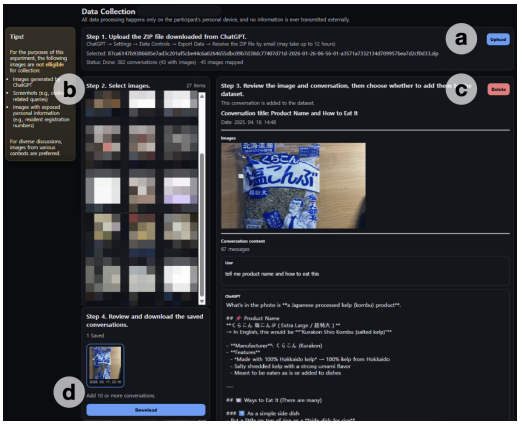


Figure 12: A locally hosted web viewer that lets participants collect their personal VLM conversation data. (a) Importing conversation histories from everyday VLM services. (b) Browsing cases with image-based queries. (c) Inspecting individual conversations. (d) Selecting and exporting cases for the workshop.

paper for the segment's response–scene relation and content (e.g., a described motion warrants a 3D arrow, and a described place warrants a panoramic view).

- **Spatial highlight.** The highlight must cover the described physical target(s). Partial coverage of a described set or a highlight on a different object counts as incorrect.
- **3D arrow.** The arrow must depict a meaningful, scene-grounded motion or direction, with start and end points aligned to the described trajectory. An arrow that terminates in midair or contradicts its paired highlight counts as incorrect.
- **Reference image, panoramic view, and 3D object.** The visual must depict the specified missing entity or target state in the described context. A generic image of the right category that misses the described aspect counts as incorrect.
- **Keywords.** Keywords must preserve the actual unfamiliar term, proper name, number, or value needed to follow the narration, in the user's language. Omitting a retention-critical term or substituting a less informative one counts as incorrect.
- **Audio Modality Generator (SSML).** Timing marks must occur at the first relevant mention and correspond one-to-one with the displayed cues, in the correct order. Sequences over multiple visible targets need sufficient transitions, and an Absent reference cue requires the specified viewing pause.

D COMPREHENSION QUIZ FOR USER EVALUATION

This section presents the comprehension quiz used to objectively assess participants' retention of information after each task. We describe the prompt used to generate quizzes and provide an example quiz from the Beverage Crafting task.

D.1 Quiz Generation Prompt

After each task, the system generated a 10-item comprehension quiz from the agent's response. We fed the user's original query and the full response text into Gemini 3 Flash[4] with the following instructions.

```
You are a quiz generator for a user study evaluating an XR assistant system. Given the user's original question and the assistant's response, generate exactly 10 multiple-choice questions to test comprehension of the response.

Each question must have exactly 4 choices (A–D) with one correct answer. Questions should vary in difficulty, cover different parts of the response, and use plausible distractors. Keep each question under 150 characters.

Input: User's question and the assistant's full response text.
Output: JSON array of 10 {question, choices, correct_answer} objects.
```

D.2 Example Quiz

Table 4 shows the quiz generated from one participant's session in the Beverage Crafting task.

Table 4: Example quiz generated for the Beverage Crafting task (pour-over coffee preparation, Set A). The quiz was generated from P04's session in which the participant asked, "Can you explain how to make a cup of coffee?".

#	Question	Ans.
1	What is the purpose of transferring boiling water into the stainless steel drip pot?	To help control the pour
2	How many grams of coffee beans are recommended for this brewing method?	15–20 grams
3	What consistency should the coffee grounds have after using the hand grinder?	Medium-coarse like sea salt
4	What should be done after placing the dripper and carafe on the digital scale?	Tare it to zero
5	How much water should be used for the initial "bloom" step?	40 ml
6	How long should you wait for the coffee to bloom before the final pour?	30 seconds
7	What is the total target volume of water for the entire brewing process?	250–300 ml
8	In what motion should the water be poured over the coffee grounds?	Steady, concentric circles
9	What type of dripper is mentioned in the instructions?	Mesh dripper
10	What is the primary purpose of the "bloom" phase in coffee brewing?	To release gases

[4] https://deepmind.google/models/gemini/flash/

Table 5: Full statistical results of the user study. Values are participant-level means across the two tasks per condition ($N = 12$). W is the Wilcoxon signed-rank statistic, p_{Holm} the Holm-adjusted p-value, and r the matched-pairs rank-biserial correlation. The two primary outcomes were corrected as one family, and exploratory measures were corrected within each scale family.

Measure	Vista M (SD)	Baseline M (SD)	W	p	p_{Holm}	r
Primary outcomes						
Comprehension quiz accuracy	6.92 (1.93)	5.75 (2.01)	11.0	0.048	0.048	+0.67
NASA-TLX overall workload	3.14 (0.83)	4.04 (0.79)	9.0	0.016	0.032	−0.77
Perceived knowledge gain						
Gained new information	6.25 (0.78)	4.92 (0.85)	0.0	$< .001$	0.002	+1.00
Improved understanding	5.96 (0.96)	4.67 (1.07)	1.0	0.004	0.008	+0.96
Thought more deeply	5.17 (1.11)	3.79 (1.08)	4.0	0.008	0.008	+0.88
Perceived information quality						
Relevance	6.04 (0.54)	5.62 (1.00)	8.5	0.125	0.125	+0.62
Detail	6.42 (0.67)	5.67 (0.83)	2.5	0.010	0.020	+0.91
Clarity	5.62 (1.17)	3.75 (1.47)	1.5	0.003	0.009	+0.95
NASA-TLX subscales						
Mental demand	3.08 (1.43)	4.96 (1.32)	0.0	$< .001$	0.006	−1.00
Physical demand	2.75 (1.66)	3.12 (1.69)	13.0	0.508	0.508	−0.28
Temporal demand	2.50 (1.55)	4.58 (1.33)	4.0	0.007	0.021	−0.88
Performance	5.25 (1.20)	3.21 (1.36)	0.0	$< .001$	0.006	+1.00
Effort	3.00 (1.13)	4.33 (1.09)	4.5	0.031	0.062	−0.80
Frustration	2.25 (1.14)	4.04 (1.36)	5.0	0.005	0.020	−0.87

E FULL STATISTICAL RESULTS OF THE USER STUDY

Table 5 reports the full test statistics for the comprehension quiz and all questionnaire measures in the user study.